



9

Time

Tune In

Go to MyInspire for



The life cycle of a butterfly is truly amazing. Butterflies have four life stages. Each of the four stages is unique to individual species of butterflies, which is part of what makes watching and raising butterflies so much fun. The names of the four stages are given below.

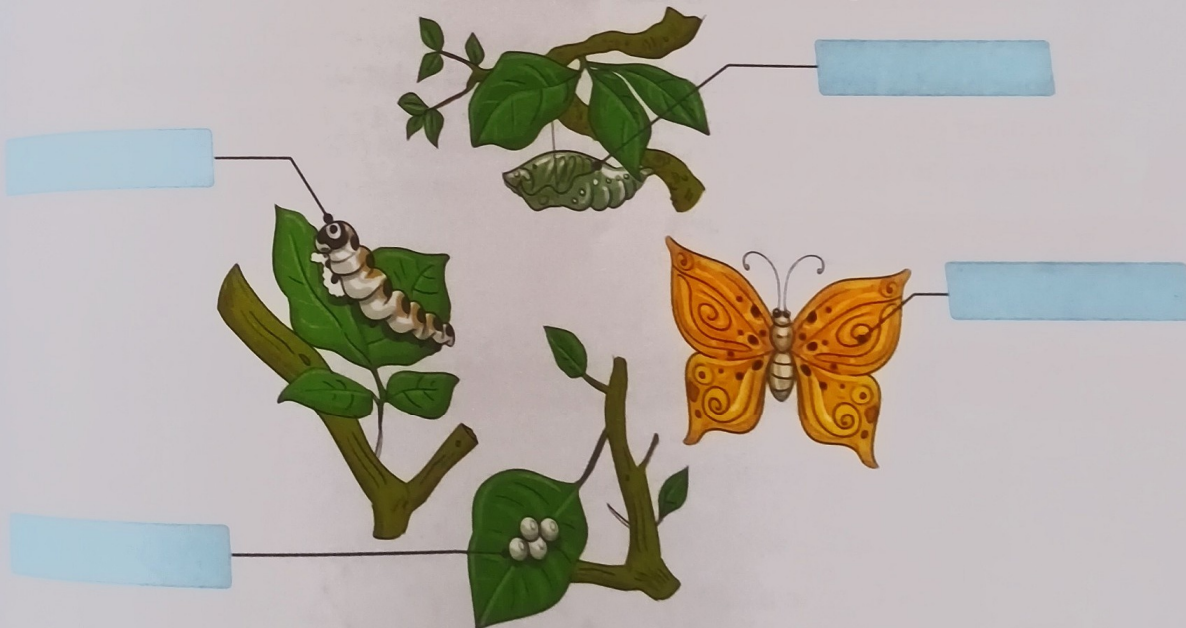
Eggs

Chrysalis

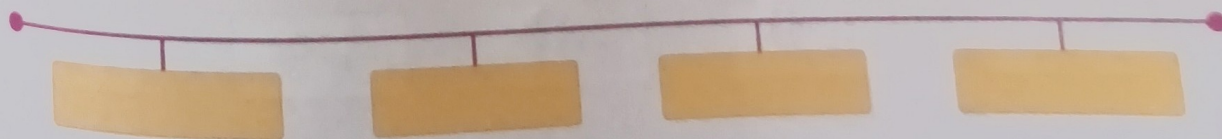
Adult butterfly

Caterpillar

a. Write the names of the stages of a butterfly in the correct blank spaces.



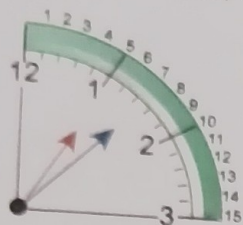
b. Plot the names of the stages of the formation of a butterfly in the correct sequence on the below timeline.



TELLING TIME TO THE MINUTE

We already know how to tell the time when the minute hand is at one of the numbers from 1 to 12.

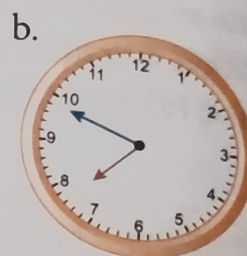
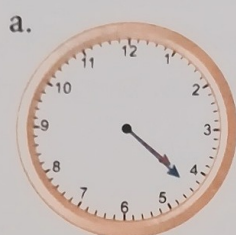
Now, suppose that the minute hand is at one of the marks between two numbers, how will we find the time?



Remember, it takes 5 minutes for the minute hand to move from one number to the next.

The interval between two numbers is divided into 5 equal parts. Each part is 1 minute.

Example 1: Find the time shown by the clocks.



Solution:

- The hour hand is just beyond 4. The minute hand is 2 small units beyond 4.
The number of minutes after the hour = $4 \times 5 + 2 = 20 + 2 = 22$ min.
The time on the clock is 4:22.
- The hour hand is between 7 and 8. The minute hand is 4 small units beyond 9.
The number of minutes after the hour = $9 \times 5 + 4 = 45 + 4 = 49$ min.
The time on the clock is 7:49.

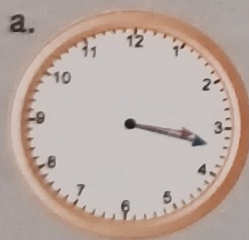
Go to MyInspire for



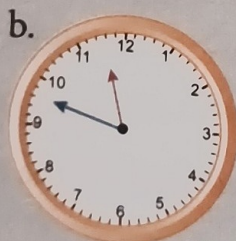
Exercise 9A

Go to MyInspire and attempt question 1.

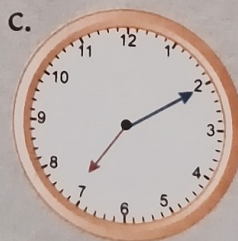
2. Write the time shown by the following clocks.



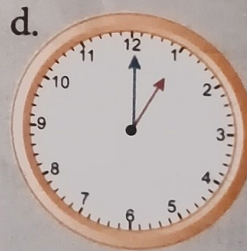
3:17



11:47



2:35



1:00



6:32

3. The clo
a.

Applica

4. Ram s
on a c

12-HC

There are

a.m.—an

p.m.—po

The 12 h

noon to

12:
mid

Exampl

a. 7:15

b. 5:30

c. 10:2

d. 1:40

Solutio



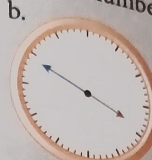
from one

rts.

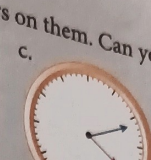
3. The clocks below have no numbers on them. Can you tell the time?



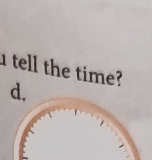
2:30



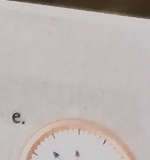
3:50



5:10



8:20



11:55

Application Based Question

4. Ram sat down to do his homework at 22 minutes past 6 in the evening. Show this time on a clock.

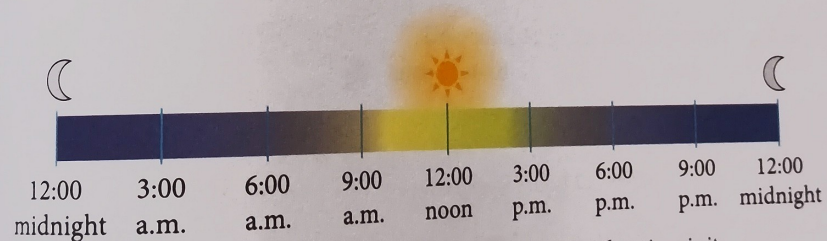
12-HOUR CLOCK

There are 24 hours in a day and 12 noon divides a day into two time periods.

a.m.—ante meridian (means before midday)—before noon

p.m.—post meridian (means after midday)—afternoon

The 12 hours from midnight to noon are the a.m. hours and the other 12 hours from noon to midnight are the p.m. hours.



Example 2: Write the time using a.m. or p.m. | **Example 3:** What time is it

a. 7:15 in the morning.

b. 5:30 in the evening.

c. 10:22 at night.

d. 1:40 at night.

Solution: a. 7:15 a.m. b. 5:30 p.m.
c. 10:22 p.m. d. 1:40 a.m.

a. 1 hour after 12 noon?

b. 1 hour before 3:30 p.m.?

Solution:

a. 1 hour after 12 noon will be 1:00 p.m.

b. 1 hour before 3:30 p.m. will be 2:30 p.m.

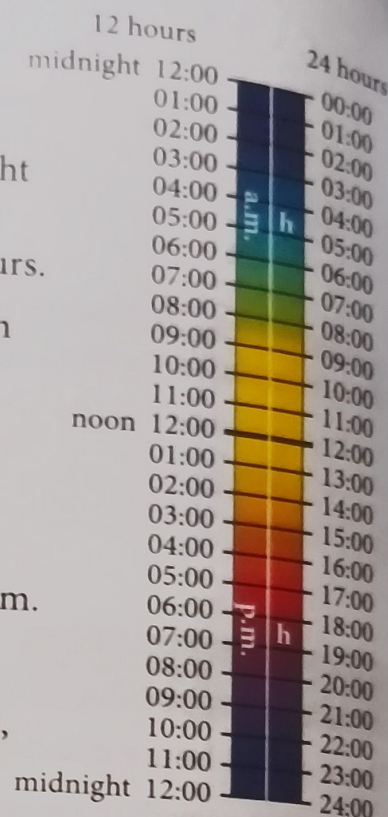
24-HOUR CLOCK

In the 24-hour clock, the time interval from midnight to the next midnight is divided into 24 hours, counted from 0 to 24.

- The 24-hour notation is in the form hh:mm—the first two digits stand for hours and the next two for minutes. Midnight is 00:00 h and noon is 12:00 h. 24:00 hours also refers to the midnight of the following day and is again read as 00:00 hours.
- From 00:00 to 12:00 hours on the 24-hour clock, the time on the 12-hour clock is the same time in a.m.
- From 12:00 to 24:00 hours on the 24-hour clock, the time in p.m. is got by subtracting 12.

12 is subtracted from the hours if the time is greater than 12 to obtain the time on a 12-hour clock. 07:40 hours refers to 7:40 a.m. and 17:40 hours refers to 5:40 p.m. in the 12-hour clock.

The 24-hour clock is used in railway timetables, flight schedules, television, and radio programme schedules.



Conversion of Time Between the 12-Hour Clock and the 24-Hour Clock

Let us learn how to convert the time from a 12-hour clock to a 24-hour clock and from a 24-hour clock to a 12-hour clock.

Example 4: Write the time using 24-hour clock.

- a. 5:00 a.m. b. 6:25 p.m.

Solution:

- a. 05:00 (the time is the same)
b. $6:25 + 12 = 18:25$ hours (12 hours is added)

Example 5: Write the time using 12-hour clock.

- a. 09:45 hours b. 16:15 hours

Solution:

- a. 9:45 a.m. (the time is the same)
b. $16:15 - 12 = 4:15$ p.m. (12 hours is subtracted)

Let's Try! 1

- Write using 24-hour clock.
a. 3:35 a.m. b. 4:25 p.m.
- Write using 12-hour clock.
a. 05:35 hours b. 17:50 hours

Write the following using 24-hour clock.

- a. 7:10 p.m. b. 10:55 a.m. c. 12:05 a.m. d. 12:30 p.m.
e. 6:40 a.m. f. 1:22 p.m. g. 9:00 p.m. h. 11:15 a.m.

Write the following using 12-hour clock.

- a. 05:45 hours b. 22:17 hours c. 00:10 hours d. 16:22 hours
e. 12:28 hours f. 13:45 hours g. 09:24 hours h. 02:05 hours

Complete the following table.

	Time in words	24-h clock	12-h clock
a.	7 o'clock in the evening	19:00	7:00
b.	two-twenty pm	14:20 hours	2:20
c.			12 noon
d.			7:10 p.m.
e.	midnight		
f.		02:45 hours	
g.	20 minutes to 2 in the afternoon		
h.	17 minutes past 4 in the morning		
i.			10:15 p.m.

6. Rewrite the following using the 24-hour clock.

- a. The play started at 6:30 p.m. b. Mr Roy goes to office at 8:45 a.m.
c. Raghav comes from school at 2:10 p.m. d. The school bus comes at 7:10 a.m.
e. Riya eats her dinner at 8:15 p.m.

Application Based Question

7. John and Stephan agreed to meet at the book store. John arrived at 4:47 p.m. and Stephan arrived at 15:45. Who arrived first?

CALCULATING TIME

Finding Time Duration

The duration of a particular activity is the time between its start and its end.

Example 6: Rihana decides to work at an old-age home to help older people. She reaches there to help the older adults at 5:20 p.m. and leaves at 6:15 p.m. How long does she stay at the old-age home? Which value does Rihana depict?

Solution: 5:20 p.m. $\xrightarrow{40 \text{ minutes}}$ 6:00 p.m. $\xrightarrow{15 \text{ minutes}}$ 6:15 p.m.

40 minutes + 15 minutes = 55 minutes

Rihana spent 55 minutes in the old-age home.

Value: 'Caring for elderly and needy people.'

Example 7: Rohan's school begins at 7:50 a.m. and gets over at 1:30 p.m. How long does the school works?

Solution:

7:50 a.m. $\xrightarrow{10 \text{ minutes}}$ 8:00 a.m. $\xrightarrow{4 \text{ hours}}$ 12:00 noon $\xrightarrow{1 \text{ hour}}$ 1:00 p.m. $\xrightarrow{30 \text{ minutes}}$ 1:30 p.m.

10 minutes + 4 hours + 1 hour + 30 minutes
= 5 hours 40 minutes

Rohan's school works for 5 hours 40 minutes.

Finding Finishing Time

If we know the starting time and the duration of an activity, we can find the finishing time of the activity.

Example 8: Kamil started working on homework at 5:45 p.m. It took him 1 hour and 30 minutes to complete it. At what time did he complete his homework?

Solution: 1 hour after 5:45 p.m. is 6:45 p.m. and 30 minutes after 6:45 p.m. is 7:15 p.m.

5:45 p.m. $\xrightarrow{1 \text{ hour}}$ 6:45 p.m. $\xrightarrow{30 \text{ minutes}}$ 7:15 p.m.

Kamil finished his homework at 7:15 p.m.

Example 9: India's women's national cricket team is also known as Women in Blue or Team India. It is governed by Board of Control for Cricket in India. Ruhi is inspired by Priya Punia, a young talent and Indian cricket captain. So she started taking coaching for cricket. The coaching started at 11:30 a.m. and lasted for 2 hours and 15 minutes. At what time did the coaching get over?



Let's Try! 2

1. Find the time duration between 2:20 a.m. and 3:15 a.m.
2. Find the time duration between 11:30 a.m. and 5:40 p.m.

Solution:
and 15 m
11:30 a.m.
The coach
Go to
Exercis
2. Find
a. 10
c. 5
3. Find
a. 2
c. 1
Applic
4. A vic
the c
5. The
It lar
was
6. An c
ever
7. Gop
Gop
to c
did
8. A tr
At v
9. Loc
the
10. The
star

Solution: 2 hours after 11:30 a.m. is 1:30 p.m.
 and 15 minutes after 1:30 p.m. is 1:45 p.m.
 11:30 a.m. $\xrightarrow{2 \text{ hours}}$ 1:30 p.m. $\xrightarrow{15 \text{ minutes}}$ 1:45 p.m.
 The coaching got over at 1:45 p.m.

Go to MyInspire for

Key Points

- Change a.m. to p.m. when you cross 12 noon.
- Change p.m. to a.m. when you cross 12 midnight.

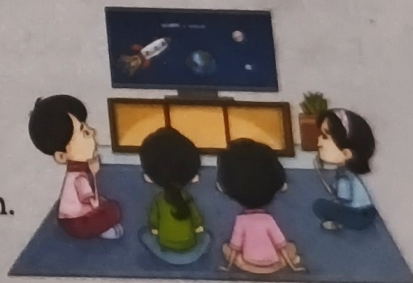
Exercise 9C

Go to MyInspire and attempt question 1.

- Find the time duration.
 - 10:10 a.m. to 11:00 a.m.
 - 8:15 a.m. to 2:30 p.m.
 - 5:20 p.m. to 9:50 p.m.
 - 9:35 p.m. to 1:20 a.m.
- Find the finishing time.
 - 25 minutes after 6:45 a.m.
 - 50 minutes after 2:50 p.m.
 - 1 hour after 9:20 a.m.
 - 4 hours 15 minutes after 10:30 a.m.

Application Based Questions

- A violin concert began at 5:15 p.m. and ended at 9:00 p.m. What was the duration of the concert?
- The flight from Chennai to Dubai took off at 04:45 hours. It landed in Dubai at 09:15 hours Indian time. How long was the flight?
- An educational show is telecast from 8:30 p.m. to 9:15 p.m. every evening. What is the duration of the show?
- Gopal has decided to help his parents to run the house. Gopal started helping her father in cooking food at 10:20 a.m. It took him 50 minutes to cook the food and he also spent 10 minutes cleaning up the kitchen. When did he finish the work?
- A train leaves New Delhi at 6:45 a.m. and takes 4 hours 20 minutes to reach Jaipur. At what time does it reach Jaipur?
- Look at the smartwatch on the right. It is 20 minutes slow. What is the correct time? Write your answer according to the 12-hour clock.
- The school's annual day function was 2 hours 40 minutes long. If it started at 10:30 a.m., at what time did it end?



CALENDAR

Let us recall what we have learnt in the previous classes.

24 hours make a day and 7 days make a week.

An ordinary year has 365 days and a leap year has 366 days, which comes once in four years.

Let us see how to find the number of weeks in any ordinary year and a leap year.

Number of Weeks in the Year

We know that a week has 7 days.

So, to find the number of weeks in a year we will divide the number of days in a year by 7.

Ordinary Year	
52 weeks	
7	365
-	35
	15
-	14
	1 day

Leap Year	
52 weeks	
7	366
-	35
	16
-	14
	2 days

365 ÷ 7 gives 52 as the quotient and 1 as the remainder.

So, an ordinary year has 52 weeks and 1 day.

366 ÷ 7 gives 52 as the quotient and 2 as the remainder.

So, a leap year has 52 weeks and 2 days.

Months of the Year

All the 12 months do not have the same number of days.

Months of the year			
January (31 days)	February (28 or 29 days)	March (31 days)	April (30 days)
May (31 days)	June (30 days)	July (31 days)	August (31 days)
September (30 days)	October (31 days)	November (30 days)	December (31 days)

Need of a Leap Year

According to the Gregorian calendar, a year has 365 days, which is the time taken by the Earth to revolve once around the Sun. But actually the Earth takes $365\frac{1}{4}$ days to revolve around the Sun.

We cannot have $365\frac{1}{4}$ days in a year.

As, 4 quarters = 1, ($\frac{1}{4}$ of 4 = 1) one full day is added to every fourth year, which will have 366 days. This year is considered as a leap year.

so, a leap year occurs once in 4 years.

If we take the total number of days in 4 years, we can see that

$$365 \frac{1}{4} \times 4 = 365 \times 4 + \frac{1}{4} \times 4 = 1460 + 1 = 1461 \text{ which is the same as,}$$

$$365 + 365 + 365 + 366 = 1461.$$

ESTIMATING TIME

Estimated time is the approximate time taken for the happening of an activity or an event. Activities or events might take seconds, minutes, hours, days, weeks, months, or even years. Here are some of the examples.

It takes minutes to eat lunch.

It takes hours for a sports day event.

It takes days to paint a house.

- It takes weeks to finish a project.

- It takes months or years to build a house.

- It takes years to complete schooling.

CALCULATING DAYS

Some activities or tasks might take days to be completed. Let us see some example to calculate the number of days between two dates.

Example 10: Find the number of days from 1st January to 2nd February.

Solution: Number of days in January = 31 days (From 1st January to 31st January)

Number of days in February = 2 days (1st and 2nd February)

Total number of days = $31 + 2 = 33$ days

So, there are 33 days from 1st January to 2nd February.

Example 11: Varun started his assignment on 21st March and completed it on 1st April.

How many days did Varun complete his assignment?

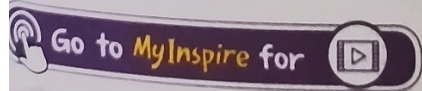
Solution: March has 31 days and Varun started his assignment on 21st March.

Number of days in March = $31 - 20 = 11$ days (Since 21st March is included)

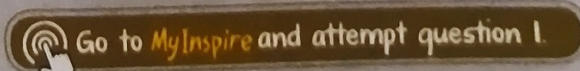
Number of days in April = 1 day

Total number of days = $11 + 1 = 12$ days

So, Varun completed his assignment in 12 days.



Exercise 9D



Calculate the number of days. Include both dates.

a. From 15th March to 13th April.

b. From 12th November to 12th December.

c. From 8th August to 18th October.

d. From 20th April to 1st June.

Application Based Questions

3. Estimate the time taken for each activity.

a.



Eating an ice-cream

5 m

b.



Visiting a beach

2 h

c.



Painting a house

1.5 h

d.



Buying tickets

5 m

e.



Doing homework

30 m

f.



Stitching a dress

5 m

- Bhanu was on leave from 26th August to 12th September. Find the number of days Bhanu was on leave.
- A weaver started weaving a saree on 1st June and completed the task on 10th July. How many days did the weaver take to complete the saree?
- Parteek enrolled for the summer camp from 7th May to 15th June. How long was the summer camp?
- The school closes for winter vacation from 25th December to 15th January. How many days will the school remain closed?
- The year 2020 is a leap year. What are the three leap years after 2020?

TIMELINE

You have already learnt about timeline in the previous classes.

We can make a timeline to show a sequence of events occurring through a day by making a 24-hour timeline. We can also make timelines representing events which occurred over a span of many years.

Example 12: Given below are some important events from the Indian history. Represent the events in their order of occurrence using a timeline.



1947
India got independence

1952
First general elections held in India

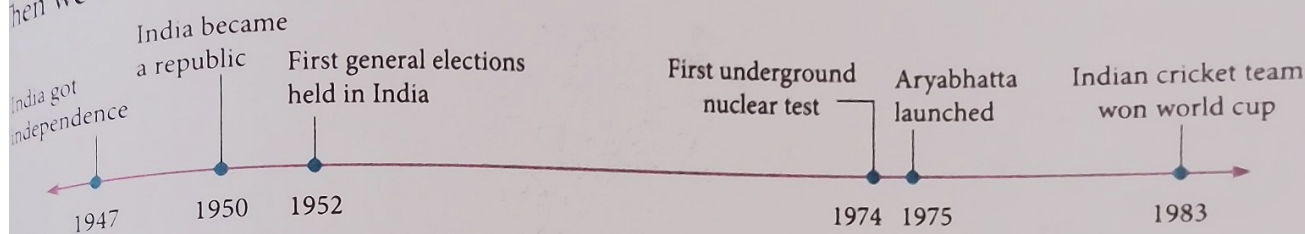
1974
India conducted its first underground nuclear test

1983
Indian Cricket team won the world cup

1950
India became a republic

1975
Aryabhata, first Indian satellite launched

Solution: First, we will arrange the years in increasing order and then plot them on a horizontal timeline. Then we will label each year with the event it is associated with.

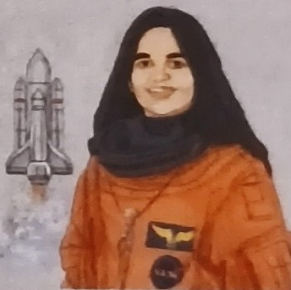


In a horizontal timeline, time is written below the line and the events are written above the line.



Exercise 9E

Kalpana Chawla was an India-born American astronaut. She was the first Indian-born woman went to space. The below information is about her life and achievements. The information is all mixed up.



1997

First Journey into space

2003

Second journey into space

1995

Started working with Nasa

1962

Born in Karnal, Hisar

1982

Moved to the US to complete her education

2001

Selected for the second journey into the space.

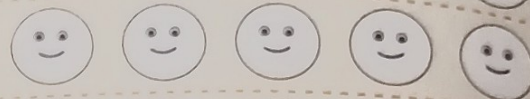
Draw a timeline and plot the years on a timeline along with Kalpana Chawla's age at the time and the landmark events.

Come here after you have completed the chapter. Colour 1 to 5  to show how well you understand a topic. (5  = Very well; 1  = Not at all)

Telling Time to the Minute



12-Hour and 24-Hour Clock



Calculating Time



Calendar



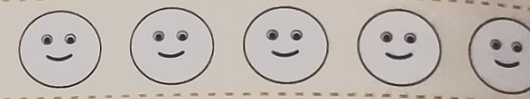
Estimating Time



Calculating Days



Timeline



Revision Exercises



Go to MyInspire for



Start Up



1. Fill in the blanks with a.m. or p.m.

- The teacher went to school at 7:15 _____.
- Rashmi went out for shopping with her friends at 6:30 _____.
- Salma's mother packed lunchbox for her at 6:45 _____.
- Rohan had his dinner with his family at 8:30 _____.

2. Complete the table.

	Time	12-hour clock	24-hour clock
a.	10 past 6 in the morning		
b.	Quarter to 3 in the evening		
c.	Half past 6 in the evening		
d.	12 noon		

c.	Half past 12 in the afternoon		
f.	Five past 8 in the evening		
g.	Twelve to 6 in the morning		
h.	17 min past 4 in the morning		

3. Fill in the empty boxes.

	Start time	End time	Time taken		Start time	End time	Time taken
a.	6:00 hours	21:25 hours		b.	04:30 hours		3 h 45 min
c.	00:00 hours		12 h	d.	05:40 hours	12:00 hours	
e.	1:45 hours	17:00 hours		f.	12:00 hours		14 h 40 min
g.	15:15 hours	23:00 hours		h.	21:25 hours	06:00 hours	
i.		11:00 a.m.	6 h 30 min	j.	6:25 p.m.		45 min
k.	12:30 a.m.	12:00 noon		l.	6:00 a.m.	8:15 a.m.	
m.	7:45 a.m.		5 h 45 min	n.	12:00 noon		24 h
o.	4:30 p.m.		10 h	p.	2:30 p.m.		3 h 30 min

Level Up



Solve the following word problems.

- Rohan started working on his project at 10:20 a.m. He worked for 3 hours 50 min when his mother called him for lunch. What time was it?
- Kumar and his friends were watching a cartoon film. The film began at 4:40 p.m. and ended at 6:20 p.m. They took a break for 15 minutes in the middle to have tea. For how long did they watch the film?
- Summer vacation is the best time for children to visit their native villages, hill stations, resorts, and religious places. Rubina decided to go to Darjeeling to visit Padmaja Naidu Himalayan Zoological Park. This is the only specialised zoo in the nation for its conservation breeding programmes for Red Pandas, Snow Leopards, Tibetan Wolves, and other critically endangered animal species of the Eastern Himalayas. So she went there with her family and saw many different type of animals. They all were there from 9:25 a.m. to 2:30 p.m. How much time did Rubina spend in the park?
- A train left Chennai at 07:30 hours on a certain day and reached Bengaluru at 14:10 hours. How long did the train take to complete the journey?
- The school closes for summer vacation from 11th May to 25th June. How many days will the school remain closed?
- Mr Sharma reached Agra on 26th September and returned home on the morning of 3rd October. How long did he stay in Agra?



7. Every evening at 5:25 p.m., Rahul goes to play cricket with his friends in a neighbouring park. He plays for 1 hour 20 minutes and takes 3 minutes to reach home. One day while returning home, he noticed a puppy lying on the side of the road, terribly injured. He immediately called the animal rescue team, who saved the puppy's life. After that, Rahul brought dog food and milk to the puppy with the help of his friends. Because of this, he was late and reached home at 7:50 p.m. He described the whole scenario to his parents, who were proud of him.



- At what time does Rahul come back from the park after playing?
- By how many hours or minutes did Rahul arrive home late?

HOTS Questions



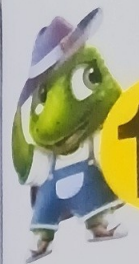
- Sally leaves her house in the morning to go to day camp. Just as she is leaving her house, she looks at the reflection of the clock in the drawing-room. There are no numbers on the clock. So, Sally makes an error in reading the time. She assumes the clock is showing 4:45. What is the correct time?
- Esther wants to sleep well for 9 hours every night. What time should she go to bed at night to wake up at 6 a.m.?

Everyday Maths



Human life revolves around time. Time helps us structure and organize our daily activities. Think and discuss the importance of your time with your friends, teachers, and parents.

- When we buy a product, we usually read the information written on its pack. It has all the important information about the product, such as its uses, ingredients, etc. It also has the 'date of manufacturing' and within how many days or months we should use the product. The number of days within which the item should be used, is also called its shelf life.
 - The date of manufacturing printed on a pack of biscuits is 25th May 2020 and it should be used before 10th September 2020. Find the shelf life of the biscuits.
 - The date of manufacturing printed on a pack of butter is 16th September 2020 and it has a stamp which says "Use by 31st October 2020". Should we use the butter after 31st October 2020? Write 'Yes' or 'No'.
- Visit a zoo along with your parents or teacher. Make a list of the animals you see. Find out the cost of each ticket. Also, find out the opening and closing times of the zoo. Note down your entry and exit time. Find out the total money spent on the tickets. Also, find out the time you spent in the zoo. Draw a timeline for this field trip.



Tune In

Go to My

Observe the
sold by a sh

24 sets sold

Total amount



Adult
Price per

Date:

S.No. N

Total

Based on the

Ravi has be
the right fo
a pair of sh
money to b